

ATUL AGARWAL

Blockchain & Backend Engineer • Goa, India (Remote, any timezone) • atulagarwal893@gmail.com • atulagarwal.dev

SUMMARY

Blockchain and backend engineer with 5+ years building financial protocols. Co-founded a perpetual futures exchange (Avalanche/Solidity) and a leveraged lending protocol (Aptos/Move) — \$7M peak TVL. End-to-end ownership: smart contracts, liquidation engines, cross-chain bridges, security audits, and backend infra (indexers, bots, APIs). Open to protocol, smart contract, and backend engineering roles in DeFi, RWA, and financial infrastructure.

EXPERIENCE

Moar Market

2024 – 2026

Co-Founder & CTO • Aptos • Move/Rust, TypeScript

- Designed and shipped a leveraged lending protocol: collateral vaults, variable-rate interest models, oracle integrations (Pyth, Switchboard), and liquidation logic — peaked at \$7M TVL
- Defined risk parameters, collateral factors, and liquidation incentive curves for multi-asset markets
- Coordinated end-to-end security audit (scoping → triage → remediation → re-review); co-built backend infra: indexer pipeline, liquidator bot, risk monitoring, REST API

Hubble Exchange

2021 – 2024

Protocol Engineer • Avalanche • Solidity, Go, TypeScript

- Owned all smart contracts for a perpetual futures DEX across two versions: V1 (virtual AMM using CurveCrypto invariant — Newton's method solver in fixed-point Solidity) and V2 (on-chain order book on a custom Avalanche Subnet)
- Built cross-chain asset bridge (LayerZero, lock-mint pattern), margin engine, funding rate mechanism, and multi-collateral vault system
- Managed security audits across both versions; established code-review practices and mentored junior engineers on Solidity, EVM internals, and DeFi attack vectors

DefiDollar

2021

Smart Contract Engineer • Ethereum / Solidity

- Developed contracts for a stablecoin index protocol — aggregated stablecoins into a diversified, risk-adjusted token

MediaTek

2017 – 2020

Chip Design Engineer

- Digital circuit design for production SoCs — timing closure, RTL verification, and physical design across multiple tapeouts

TECHNICAL WRITING

- [“Understanding the Curve AMM: StableSwap Invariant”](#) — Newton’s method convergence, swap mechanics, and bonding curve math
- [“Hubble vAMM: CurveCrypto Invariant”](#) — extending Curve’s invariant to volatile-asset perpetual futures pricing

SKILLS

Smart Contracts: Solidity (Foundry, Hardhat), Move (Aptos)

Protocols & Finance: Lending/borrowing, perpetual futures, AMM design, cross-chain bridging (LayerZero)

Security: Audit coordination & remediation, code review (reentrancy, oracle manipulation, overflow), Foundry fuzzing

Backend & Infra: Ethereum/EVM, Avalanche Subnets, Go, Aptos/MoveVM, Rust, TypeScript, Python, MongoDB

EDUCATION

Indian Institute of Technology (IIT) Delhi • M.Tech, Control and Automation Engineering, 2017

Indian Institute of Technology (IIT) Jodhpur • B.Tech, Electrical Engineering, 2015